

Sec. 2.3: Best practices and common errors in proofs

- ★ See zyBook Figure 2.3.1 for allowed assumptions in proofs
- ★ See zyBook Figure 2.3.2 for common keywords and phrases in proofs.

Best practices in writing proofs

- **Indicate when the proof starts and ends.**

Start with:

End with:

- **Write proofs in complete sentences.**

Example of a poor-written proof for “The sum of two odd integers is an even integer. ”

Proof. $x, y \rightarrow \text{odd}$

$$x = 2k + 1$$

$$y = 2j + 1$$

$$\rightarrow x + y = 2k + 1 + 2j + 1 = 2(\underbrace{k + j + 1}_{\text{integer}}).$$

□

- **Give the reader a roadmap of what has been shown, what is assumed, and where the proof is going.**

- Start with stating the assumptions and what is to be proved.
- To use a fact that have not yet be proved, we write:
“We claim that ...”, “Claim: ...”, and prove it later in the proof.

- **Introduce each variable when the variable is used for the first time.**

- “Let x be an even integer. ”
- “Since x is even, $x = 2k$ for some integer k . ”

- **A block of equations should be introduced with English text and each step that does not follow from algebra should be justified.**

- “Then $x + y = \dots$ ”
- “Substituting $x = 2k$ into x^2 gives ...”
- “Squaring both sides of the equation yields

$$\begin{aligned}(\sqrt{2})^2 &= \left(\frac{a}{b}\right)^2 \\ 2 &= \frac{a^2}{b^2}.\end{aligned}$$

- **Check if everything in the hypothesis is actually used!**

A “good” theorem should not have redundant hypothesis that is not used in its proof. So if there is anything in the hypothesis that is not used in your proof, chance is that there is a mistake in the proof, or the statement of the theorem can be improved.

Example of a redundant hypothesis: “The product of an odd integer and an even integer is even.”

Common mistakes in proofs

- (a) Generalizing from examples.
- (b) Skipping steps.
- (c) Circular reasoning.
- (d) Assuming facts that have not yet been proven.
- (e) Invalid uses of existential instantiation (bad variable names).

Example 1. Find mistake(s) in each of the following questionable “proof”! How to fix the mistakes?

- (1) “If n^3 is even, then n is even. ”

Proof. We shall prove the contrapositive statement: “If n is odd, then n^3 is odd.”

Assume that n is an odd integer. We need to prove that n^3 is odd.

Since n is odd, $n = 2k + 1$ for some integer k .

Plugging in $n = 2k + 1$ into n^3 gives $n^3 = (2k + 1)^3$.

Therefore, n^3 is an odd integer. □

- (2) “The sum of two odd integers is even. ”

Proof. Assume that x and y are odd integers. We need to prove that $x + y$ is odd.

Since x is odd, $x = 2k + 1$ for some integer k .

Since y is odd, $y = 2k + 1$ for some integer k .

Then $x + y = 2k + 1 + 2k + 1 = 4k + 2 = 2(2k + 1)$. Since k is an integer, $2k + 1$ is also an integer.

Hence, $x + y = 2(2k + 1)$ is odd. □

- (3) “Every nonzero integer n divides 0. ”

Proof. Let n be a nonzero integer. If $n = 5$, then $0 = 0 \times 5$, which means 5 divides 0 by definition.

Therefore, every nonzero integer n divides 0. □

- (4) “The square of an even integer is even. ”

Proof. Assume that x is an even integer. We need to prove that x^2 is even.

Since x is even, $x = 2k$ for some integer k .

Since the product of two even integers is even, $x^2 = x \cdot x$ is even. □

- (5) “The square of an odd integer is odd. ”

Proof. Assume that x is an odd integer, $y = x^2$ is an odd integer. We need to prove that x^2 is odd.

Since y is odd, $y = 2k + 1$ for some integer k .

Since $x^2 = y = 2k + 1$, x^2 is an odd integer. □